CHAPTER 12

Measurement: Scaling, reliability and validity

LEARNING OBJECTIVES

After completing Chapter 12, you should be able to:

1. Describe the characteristics and power of the four types of scales – nominal, ordinal, interval, and ratio.
2. Describe and know how and when to use different forms of rating scales.
3. Describe and know how and when to use different forms of ranking scales.
4. Discuss international dimensions of scaling.
5. Describe validity and reliability and how they are established and assess the reliability and validity of a scale.
6. Explain the difference between reflective and formative scales.

INTRODUCTION

In the previous chapter, we explained that measurement is the assignment of numbers or other symbols to characteristics (or attributes) of objects according to a prespecified set of rules. Now that we have learned how to operationally define (or operationalize) a concept (or variable), we need to assign numbers (or other symbols) to it in some manner. Note that it is important that the rules for assigning numbers to characteristics of objects should be standardized and applied in a consistent manner.

Numbers allow us to perform statistical analysis on the resulting data and (in deductive research) to test the hypotheses that we have developed. What's more, they facilitate the communication of our research results.

In this chapter we will examine the types of scales that can be applied to assign numbers to characteristics of objects and subsequently see how we actually apply them. We will first discuss four different types of scales (nominal, ordinal, interval, and ratio scales) and point out that the statistical analysis we can perform later on in the research process is directly related to the type of scales we use. We will also discuss two main categories of